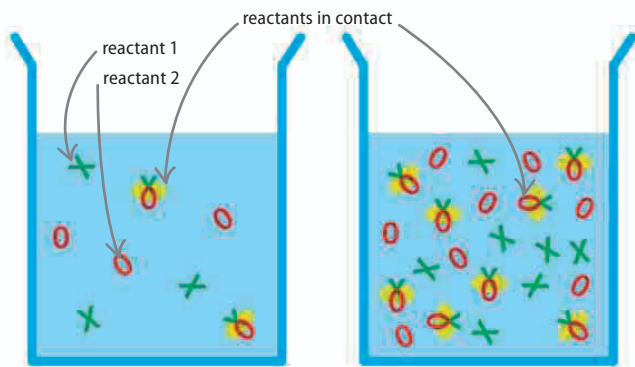


Concentration

Concentration is a measure of how much of a substance is present in a certain volume of a mixture. The rate of reaction is proportional to the concentrations of the reactants. Even if one reactant is present in large amounts, the reaction will only speed up as more of the other is added.



△ Low concentration

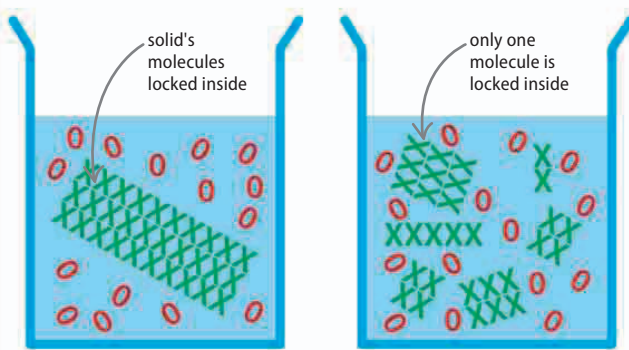
Reactants must make contact with each other to react. When reactants are mixed in low concentrations, they are widely dispersed and come into contact with each other infrequently.

△ High concentration

When reactants are mixed in higher concentrations, their molecules are less spread out and come into contact with each other more frequently. As a result, the rate of reaction is higher.

Particle size

When a solid reactant is added to a liquid or dissolved reactant, the reaction will proceed faster if the solid is crushed into powder rather than added as a single lump. The liquid reactant reacts with the surface of the solid, and the powdered reactant has a larger combined surface area than the single mass.



△ Large solid, small area

When the solid reactant is added as a big lump, the liquid reactant has fewer opportunities to react it. This is because many of the solid's molecules are locked away inside, out of reach of the liquid reactant.

△ Small solids, large area

The same amount of solid reacts much more quickly when broken up into smaller sizes. This is because more of its molecules are made available to take part in the reaction.

Light

Some reactions speed up when exposed to bright light or other higher energy forms of radiation, such as ultraviolet (UV) light. The reactants absorb the energy from certain wavelengths and this is enough to give them the activation energy to begin reacting. These reactions are called photochemical reactions. Photosynthesis, used in plants to turn carbon dioxide and water into glucose, needs light. Without it, the rate of reaction is negligible.

▷ Ozone layer

The reaction that creates ozone, a form of oxygen with three atoms per molecule happens mostly where high-energy light hits the high atmosphere. The ozone forms a layer in the high atmosphere and helps to absorb the dangerous UV rays in sunlight.

